

Scalability and Future Plan of Action

HDI Prediction System — Project Demonstration Phase

Current Scope

The HDI Prediction System currently predicts HDI using four core indicators trained on a single year (2021) of global data. The architecture is designed to scale beyond this initial scope.

Scalability Considerations

- The Flask + Render.com architecture can handle increased user traffic with minimal changes.
- The lightweight Linear Regression model keeps inference fast even as usage grows.
- Additional years of HDI data can be appended to the training set without redesigning the pipeline.

Future Enhancements

- Incorporate multiple years of HDI data to show historical trends per country.
- Add visualizations (charts/graphs) comparing predicted vs. actual HDI across countries.
- Extend the model to related indices, such as the Gender Development Index.
- Introduce user accounts to save and compare previous predictions.
- Expose a public API endpoint for programmatic access to predictions.

Long-Term Vision

Over time, the system can evolve from a single-model predictor into a broader development-analytics platform, combining multiple socio-economic indices and interactive visual insights for researchers and policy analysts.